

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

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SUBJECT: Cadillac WWTP
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257
Water Quality-Based Effluent Limits (WQBELs)

Haring Twp WWTP
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0059076
Water Quality-Based Effluent Limits (WQBELs)

Permits Section has evaluated conventional parameter water-quality based effluent limits (WQBELs) for the proposed reissuances of two NPDES permits within a waste-load allocation: the Cadillac WWTP and Haring Twp WWTP. Both facilities discharge to the Clam River in Wexford County. The two discharges are summarized in the table below.

Table 1. Waste-Load Allocation NPDES Individual Permits

Facility	Permit Number	Outfall Location	Design Flow (MGD)	Type of Wastewater
Cadillac WWTP	MI0020257	44.2657°N, -85.3985°E	3.2	Treated municipal wastewater
Haring Twp WWTP	MI0059076	44.2805°N, -85.3865°E	0.135	Treated municipal wastewater

In the 1980s, WQBELs were developed based on a trout stream designation (coldwater protection) for the Clam River between Lake Cadillac and the Wexford-Missaukee County line in the non-summer seasons. The Michigan DNR Fisheries order in 1997 and the most current fisheries order list the Clam River as being protected for coldwater fish downstream of the Wexford-Missaukee County line, and as warmwater protected between the lake and the county line.

The Clam River from Lake Cadillac to the Wexford-Missaukee county line is protected for warmwater fish species, other indigenous aquatic life and wildlife, agriculture, navigation, industrial water supply, public water supply at the point of intake, partial body contact recreation, total body contact recreation from May 1 to October 31, and fish consumption. The Clam River downstream of the county line is protected for the same designated uses in addition to being protected for coldwater fish.

The monthly exceedance flows (in cfs) for the Clam River at the Cadillac WWTP outfall are as follows based on Low Flow Discharge Request 10314:

	<u>JAN.</u>	<u>FEB.</u>	<u>MAR.</u>	<u>APR.</u>	<u>MAY</u>	<u>JUNE</u>
50%	17	17	36	78	35	13
95%	5.6	5.1	9.0	14	6.9	4.1
	<u>JULY</u>	<u>AUG.</u>	<u>SEP.</u>	<u>OCT.</u>	<u>NOV.</u>	<u>DEC.</u>
50%	7.4	6.3	6.3	10	18	19
95%	3.0	3.0	3.1	3.7	5.5	5.5

The 95% exceedance flows have increased in every month except July and August since the last low flow update. Flows are unchanged in July and August. The lowest monthly 95% exceedance flows in up to four seasons are used as the background flow of the receiving stream for the purposes of developing WQBELs. The following seasons are used, consistent with the current permit structure: June-September, October-November, December-March, and April-May.

A multiple reach Streeter-Phelps model was used to model the impact of both facilities on the Clam River at critical low-flow conditions. Effluent limit recommendations are determined for five-day carbonaceous biochemical oxygen demand (CBOD₅), ammonia (NH₃-N), and minimum effluent DO that are protective of the warmwater DO standard at all times. A diurnal variation of 0.75 mg/l per day between the daily average and daily minimum DO concentration was assumed based on a 2006 field study for the warmwater-protected stream reaches. A diurnal variation of 0.51 mg/l was assumed for the coldwater-protected reaches based on the same study. Kinetic rates were also set based on department procedure defaults for CBOD₅. A 1983 field study determined that nitrification was not occurring in the Clam River due to a lack of substrate for nitrifying bacteria. Site-specific time of passage and reaeration studies completed in 1983 were used to determine stream velocity and reaeration. Stream slopes were updated using LiDAR data. Effluent limit recommendations for the Cadillac WWTP are summarized in Table 2. Table 3 provides the effluent limit recommendations for the Haring Twp WWTP.

Effluent limit recommendations of the Cadillac WWTP in the summer season are less restrictive than current permit effluent limits due to the change in stream designation. The current permit effluent limits are based on anti-degradation in the basis of decision memo associated with the last permit reissuance. There is no daily maximum effluent limit recommendation for NH₃-N based on the assumption that nitrification is not occurring in the Clam River.

The summer season permit limits at the Haring Twp WWTP are based on federal secondary treatment standards (STS). The department conservatively assumes STS to be represented by a daily maximum CBOD₅ concentration of 60 mg/l and a daily maximum NH₃-N concentration of 30 mg/l.

Effluent limit recommendations based on federal secondary treatment standards (STS) are sufficient to protect the DO standard in the Clam River in the remaining three seasons for both facilities. A minimum effluent DO concentration of 3.0 mg/l is recommended to protect against acute toxicity in the effluent-receiving water mixing zone. The STS-based effluent limits are consistent with current permit effluent limits at the Haring Twp WWTP. The STS-based effluent limits are less restrictive than current permit effluent limits for the Cadillac WWTP.

Both discharges were also evaluated to determine 30-day average NH₃-N effluent limits that are protective of the chronic NH₃-N toxicity criterion. A new chronic toxicity criterion was adopted by the department in 2019 and is being implemented for permits being issued in the current fiscal year. An effluent pH of 7.3 SU was assumed for the Cadillac WWTP and an effluent pH of 7.2 SU was assumed for the Haring Twp WWTP based on two years of discharge monitoring data. The recommended effluent limits to protect against chronic toxicity are more restrictive than current permit effluent limits at the Cadillac WWTP in April. The limits for the remaining months at the Cadillac WWTP are equivalent to or less restrictive than current permit effluent limits.

It was necessary to add 30-day average effluent limits for the Haring Twp WWTP in the two warm weather seasons based on the new criteria.

These effluent limit recommendations are based on water quality standards. We have not addressed treatment practicality or cost effectiveness. Our recommendations do not imply that other considerations should not be taken into account when deciding on permit limits.

Table 2. Cadillac WWTP (MI0020257) WQBELs

Parameter	Months	Concentration (mg/l)	Basis	Rationale
CBOD ₅	June-September	17	Daily Maximum	DO Standard
NH ₃ -N	June-September	0.9	30-Day Average	Chronic Toxicity
	October-November	3.1	30-Day Average	Chronic Toxicity
	December-March	5.4	30-Day Average	Chronic Toxicity
	April-May	3.0	30-Day Average	Chronic Toxicity
D.O.	June-September	6.0	Minimum	DO Standard
	October-May	3.0	Minimum	Acute DO Toxicity

Design Flow = 3.2 MGD

Table 3. Haring Twp WWTP (MI0059076) WQBELs

Parameter	Months	Concentration (mg/l)	Basis	Rationale
NH3-N	June- September	13.6	30-Day Average	Chronic Toxicity
	April-May	17.2	30-Day Average	Chronic Toxicity
D.O.	June- September	4.0	Minimum	DO Standard
	October-May	3.0	Minimum	Acute DO Toxicity

Design Flow = 0.135 MGD